D06 · Network Design & 3PL Management

01 · Logistics network design: facility location & flow optimization

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Logistics network design decides how many facilities to operate, where, with what role, and how product flows between plants, DCs and customers — using optimization models that balance transport, facility and inventory cost against service.

2 WHY IT MATTERS

Network decisions lock in most of the structural logistics cost for years. **More nodes shorten delivery distance but add fixed cost and reduce risk pooling; fewer nodes do the opposite.** Centralizing inventory lowers safety stock because demand variability pools across locations **(Eppen, 1979 — Management Science)**, so network and inventory must be optimized together, not in sequence.

3 HOW IT IS DONE

1. **Build a clean baseline.** Current flows, costs and service by lane and customer, validated against the P&L.
2. **Define service targets and scenarios.** Required delivery times, growth, nearshoring shifts and disruption cases.
3. **Optimize network and inventory jointly.** Compare node counts and locations on total cost: freight + facilities + inventory.
4. **Plan the transition.** A phased path from current to future state that protects service during the move.

IN PRACTICE

Optimize network and inventory together

Sequential optimization (first network, then inventory) systematically leaves money on the table.

Always include a realistic transition plan

The optimal future-state network is useless if the company cannot get there without breaking service or incurring prohibitive change costs.

Re-evaluate the network when volume, mix or service strategy changes materially

A network designed five years ago for a different demand footprint is often no longer optimal.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — National distributor evaluating a 2-DC vs 4-DC network — Service target: 95% of demand within one-day ground transit.

Network Option	Outbound Freight	Facility + Inventory Cost	Service (% demand <=1 day)
Current (2 DCs)	Baseline	Baseline	81%
3-DC optimized	-11%	+9%	94%
4-DC optimized	-16%	+21%	97%

Result: The 3-DC design met the service target at the lowest total cost. The 4-DC option improved service only marginally while adding significant fixed and inventory cost. A phased transition plan was executed over 14 months.

HOW IT IS MEASURED

% Demand within Target Transit Time

Share of demand that can be reached within the defined service standard from the current network

Benchmark: Set by commercial strategy; network design should achieve it at minimum total cost

Total Network Logistics Cost (Freight + Facility + Inventory)

Fully loaded structural cost of the network

Benchmark: Primary objective function of network design studies

WHAT YOU WOULD USE

Network Design

Coupa Supply Chain Design (formerly LLamasoft) / anyLogistix / AIMMS — Leading Supply Chain Network Design platforms.

Blue Yonder / Kinaxis scenario capabilities — Useful for ongoing what-if analysis once the baseline network is established.

4 SO WHAT

Design the network on total cost and service, with inventory inside the model. Start from a validated baseline, test scenarios, and plan a transition the business can actually execute.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México